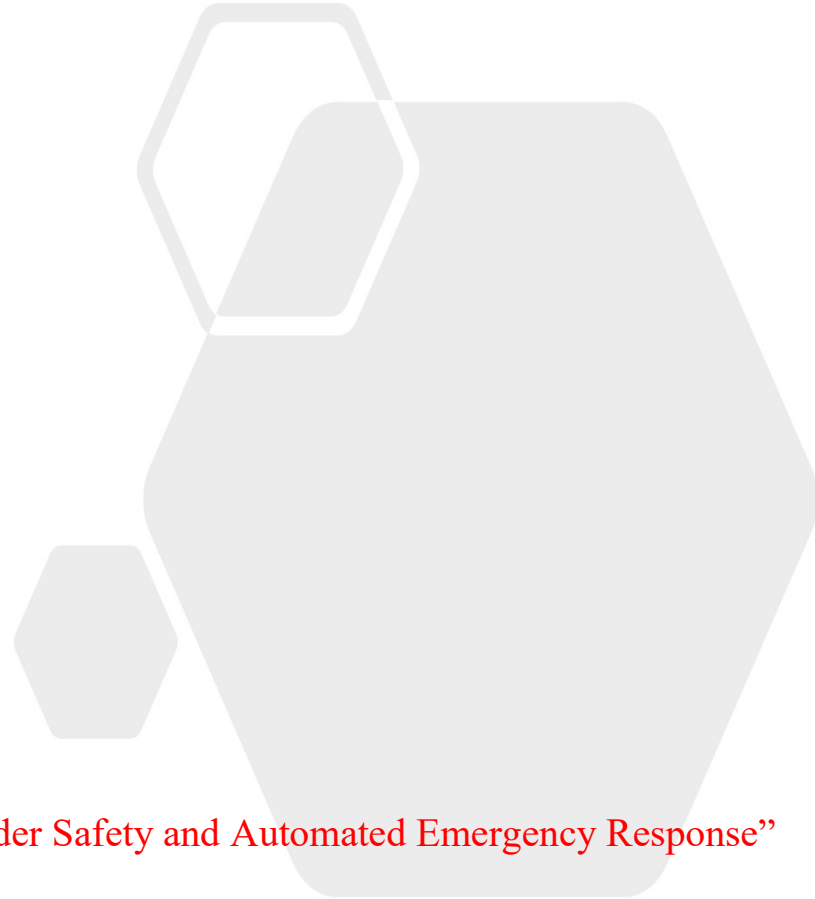


SMART HELMET — Patent Disclosure

- **SMART HELMET**
- **Inventors: Team CodeCrafters..**
- **Institution: Sharda University**
- **Year: 2025**

“An Intelligent Helmet System for Enhanced Rider Safety and Automated Emergency Response”





Invention Concept & Technical Approach

Technical Problem: Road Safety in Transportation & Logistics

A significant number of two-wheeler fatalities occur due to absence of helmet use, particularly within last-mile logistics and delivery operations. This results in:

- Operational delays and downtime in logistics networks
- Elevated insurance and compensation expenditures
- Workforce loss and rider unavailability
- Delayed medical assistance, reducing survivability and impairing transport continuity

Industry Impact Context

India records over 1.5 lakh annual road fatalities, with more than 40,000 associated with non-helmet usage. Logistics riders contribute a substantial portion to these incidents due to continuous on-road exposure and high-frequency deliveries.

Proposed Invention: Intelligent Helmet System

A smart helmet-based safety system comprising:

- A helmet-wear verification mechanism ensuring correct usage prior to vehicle operation
- Integrated sensing modules to monitor collision risk and hazardous conditions
- An automated emergency-communication feature for real-time alert transmission in the event of an accident



Novel Features & Invention Scope

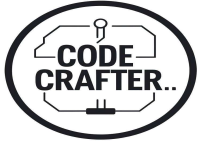
Novel Features & Invention Scope

Key inventive aspects include:

- **Ignition Interlock Mechanism**
Vehicle engine activation is prevented unless helmet usage is electronically authenticated.
- **Three-Layer Safety Architecture**
 1. Preventive hazard monitoring
 2. Collision detection and validation
 3. Mandatory helmet enforcement
- **Emergency Response Automation**
Real-time GPS-enabled emergency alert dispatch to designated medical and supervisory contacts.
- **Logistics Efficiency Enhancement**
Designed to reduce operational downtime, accelerate rescue response, and enhance workforce protection in fleet environments.

Primary Operational Advantage

The invention improves rider safety, accelerates emergency assistance, and enhances transport system reliability by enforcing protective gear compliance and automating post-incident communication workflows.



TECHNICAL APPROACH



Integrated Functional Modules:

❑ **Helmet Enforcement Module (Ignition Interlock System):**

- A force-sensitive resistor (FSR) sensor assembly verifies helmet-wear status.
- Vehicle ignition remains inhibited until valid helmet-wear confirmation is received from the sensor interface.

❑ **Collision Prevention Module (Proximity Warning System):**

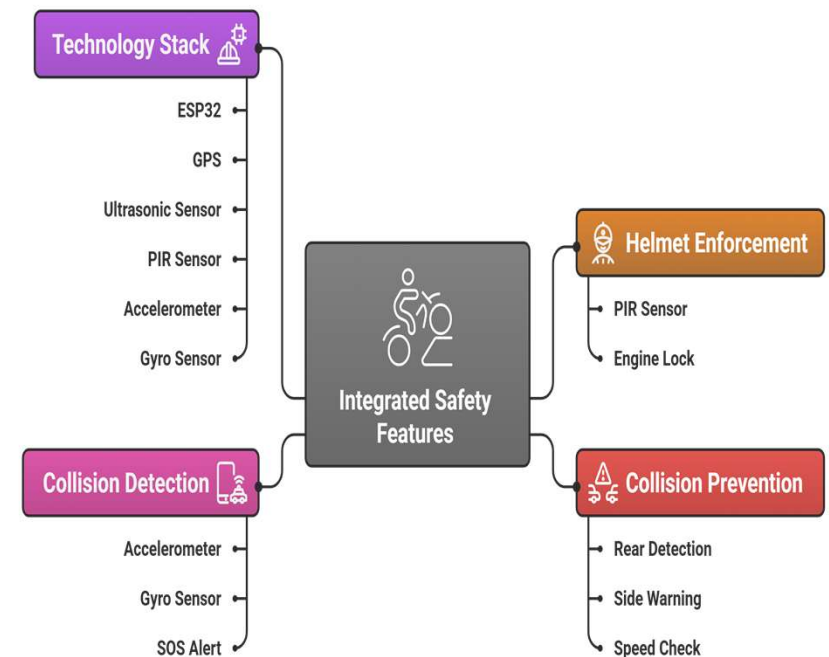
- **Ultrasonic sensing unit:** Detects rear-approaching vehicles within approximately 10 feet and generates a rider alert.
- **PIR:** Warns of objects within 4–5 ft (side).

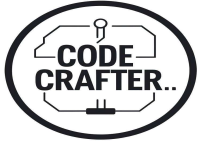
❑ **Collision Detection & Emergency Response Module:**

- A combined inertial measurement system (accelerometer + gyroscope) identifies impact events and validates crash conditions.
- GPS sends SOS with live location.
- Alerts: Police, Ambulance, Emergency Contacts.

❑ **Technology Stack:**

- **Hardware:** ESP32, GPS, Ultrasonic, PIR, Accelerometer, Gyro Sensor.





Technical Feasibility & Implementation Viability

◆ Feasibility Evaluation

•Technical Feasibility:

The system is implementable using commercially available microcontroller platforms and integrated sensing components, including inertial, proximity, and wear-detection modules.

Existing IoT technologies support the required data acquisition, processing, and wireless communication functions without specialized hardware dependency.

•**High Viability:** Long-term sustainable & effective for rider safety.

◆ Potential System Challenges

Operational Reliability Requirement

Continuous, uninterrupted system operation is essential to ensure safety performance and functional trustworthiness.

Data Security Consideration

Sensitive information, including real-time geographic coordinates and rider identity, must be protected against unauthorized access or misuse.

◆ Mitigation & Enhancement Measures

•**Security:** Use strong encryption & transparent handling.

•**Reliability:** Regular sensor checks & maintenance.





Technical Advantages & Industrial Applicability

Transportation & Logistics Benefits

- Helmet-wear detection ensures rider safety.
- Automated SOS reduces accident fatalities.
- Real-time monitoring maintains delivery continuity.
- Quick alerting enables faster rescue and traffic clearance

Industrial & Technological Advantages

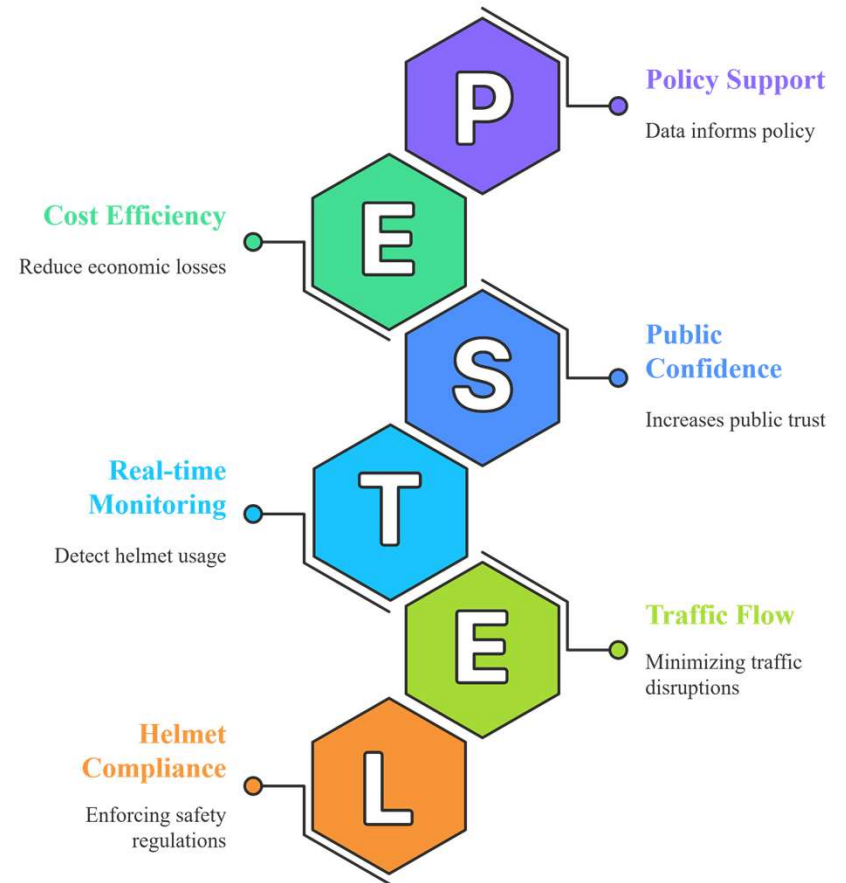
- Ensures fleet stability and rider availability.
- Provides data for predictive fleet management.
- Easily integrates with IoT and GPS systems.
- Improves overall operational efficiency.

Social & Economic Impact

- Reduces emergency response time.
- Lowers insurance and compensation costs.
- Promotes helmet awareness and safety compliance.
- Strengthens public safety standards.

Patent Significance

- Combines prevention, protection, and emergency response.
- Transforms helmets into active, intelligent safety devices.





Prior Art Review & Technical References

➤ Collision Detection & Prevention Technologies

Research Context:

Existing literature demonstrates that multi-sensor fusion architectures incorporating IoT sensing and AI-based decision models enable real-time rider-impact detection and proactive collision-avoidance functionality. These methodologies improve situational awareness and significantly reduce emergency response latency for helmet-based safety solutions.

Reference:

“Development of an AI-Integrated Smart Helmet for Enhanced Rider Safety Using Computer Vision,” *Proceedings of IEEE International Conference on Consumer Electronics-Asia*, Thailand, 2025.

➤ GPS-Enabled Rider Location Tracking

Research Context:

Studies confirm that GPS-embedded helmet systems facilitate accurate, continuous spatial tracking and route monitoring. This capability enhances rider safety, provides geolocation context during critical events, and supports emergency-assistance workflows.

Reference:

“An IoT Based Intelligent and Smart Helmet for Bike Riders Using Arduino,” *International Journal of Intelligent Systems and Applications in Engineering*, vol. 12, no. 10s, pp. 354–359, 2024.

➤ Automated Emergency Alert & Communication Systems

Research Context:

Hybrid cellular-IoT architectures integrated into helmet systems demonstrate automated incident alerting with response initiation times under ~10 seconds. These systems transmit accident information to emergency responders and verified contacts without rider intervention.

Reference:

“IoT Based Smart Helmet and Accident Identification System,” *Proceedings of IEEE Region 10 Symposium (TENSYP)*, Dhaka, Bangladesh, 2020.